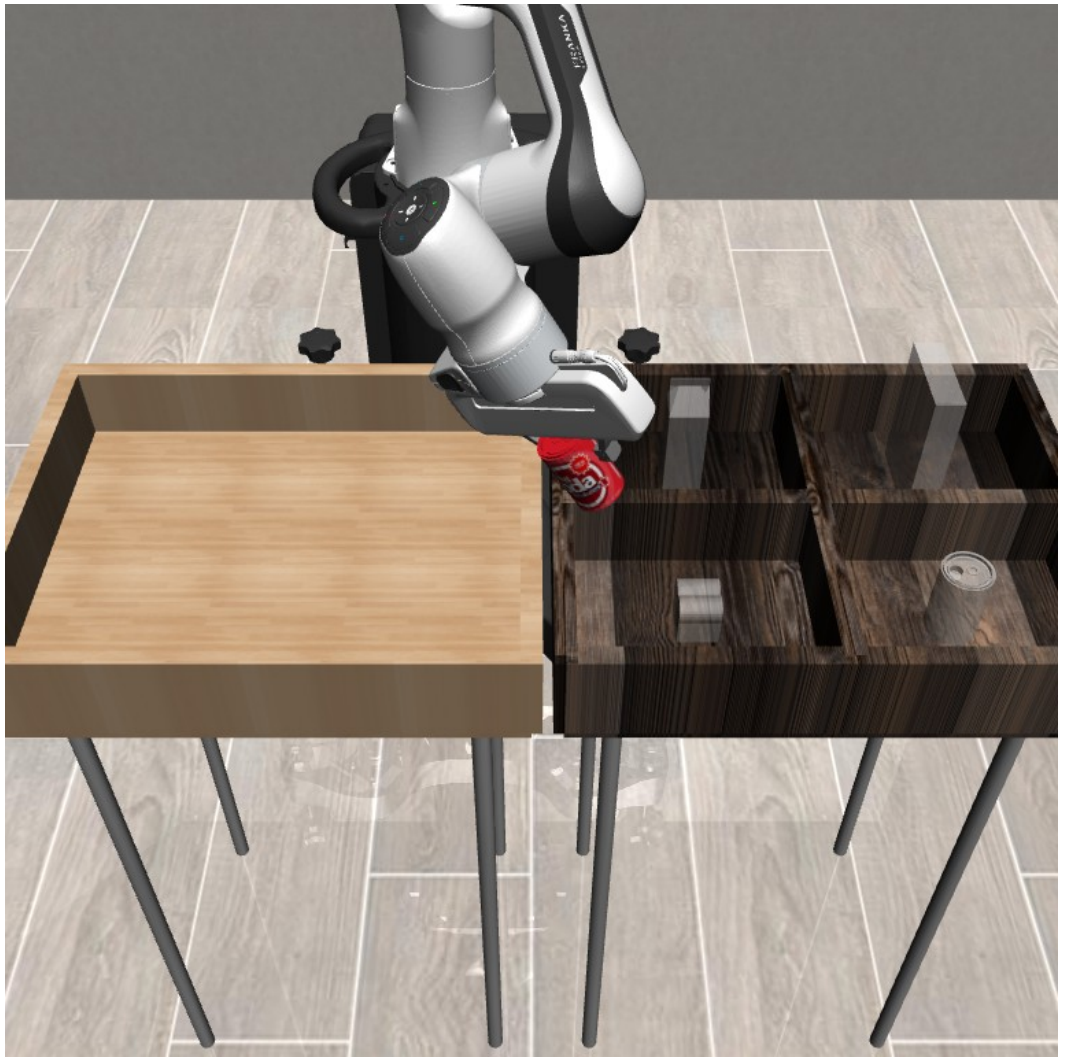
Introduction to Robotics

Manipulation Learning Lab

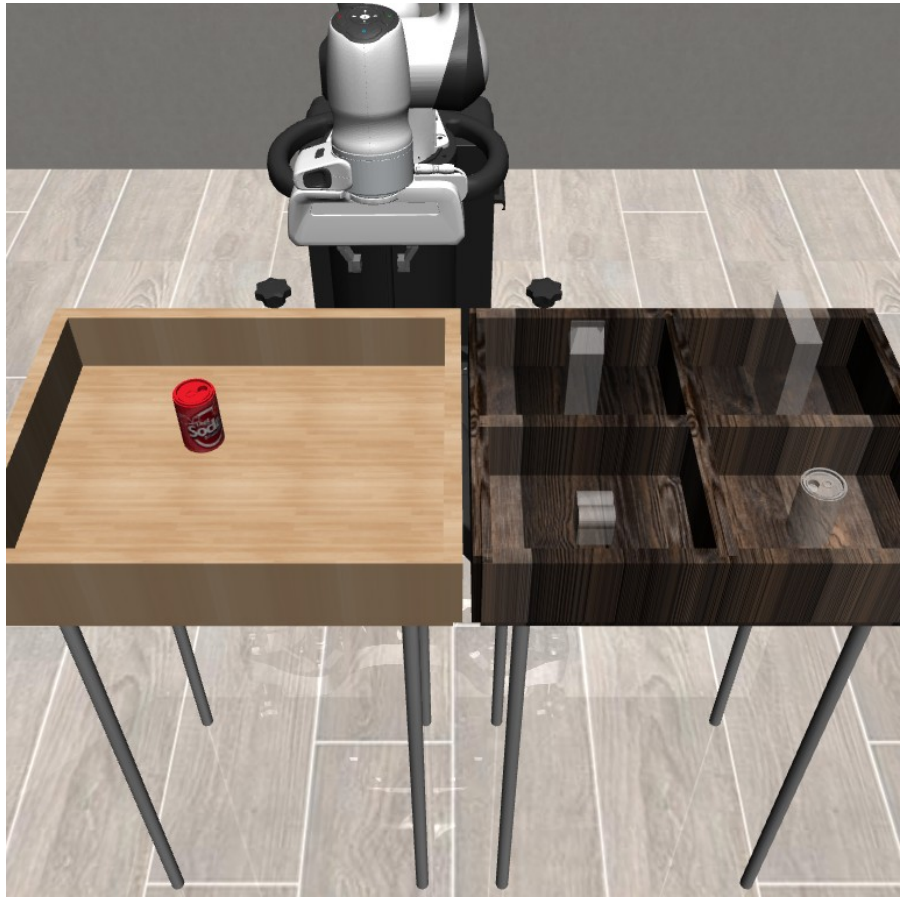
From Demonstrations to Robot Policies

Learning Robot Manipulation
from Demonstrations

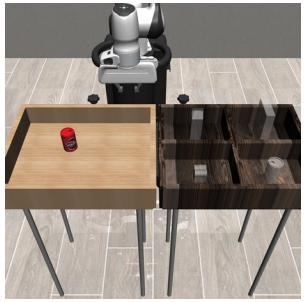


What Will the Robot Do?

Panda arm · red can · can-marked target bin



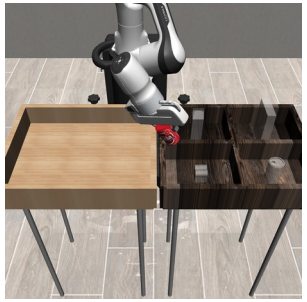
Video in PPTX · 15 s · half speed



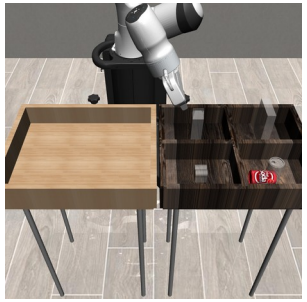
Reach



Grasp



Transport



Place

Stable released placement

Inside the target region, with a radius margin
Gripper open and no longer touching the can
Stable for 1 s after release

Displacement ≤1 cm · speed ≤5 cm/s

From demonstrations to a policy

Demonstrations → Training → Inference

Recorded observations and human actions.

Fit a model to those observation–action pairs.

Use the current observation to choose the next action.

Observe → act → observe again, at 20 Hz

200 complete demonstrations 160 train 20 validation 20 test

Select saved model weights using validation loss; reserve test cases for evaluation.

Observations and actions

Observation: 16 values

7	Gripper-tip position and orientation
2	Finger joint positions
7	Can position and orientation

World frame · Z up · positions in meters

Action: 7 values

3	Translation in X, Y, Z
3	Rotation of the gripper
1	Close (+) or open (−)

Normalized commands in [−1, 1]

$$\hat{a}_t = \pi_\theta(o_t)$$

o: normalized observation θ: learned weights

â: predicted action t: current control step

Model: 16 → 256 → 256 → 256 → 7

Training-only normalization · fixed weights during inference

Learning by behavior cloning

Match the demonstrated action, then update the model weights.

Mean squared action error

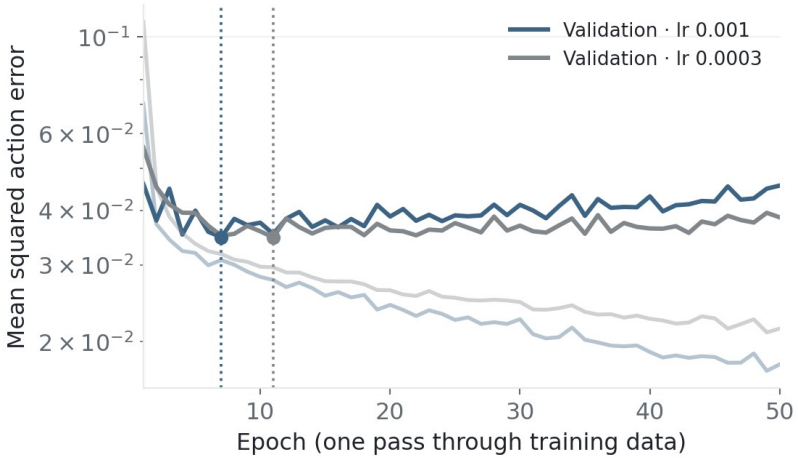
$$L_B(\theta) = \frac{1}{7B} \sum_{i=1}^{B} \left\| \hat{a}_i - a_i^E \right\|_2^2$$

B: batch size (up to 256) · i: a training example

a^E: human action · average over all 7 action values

Weight update $\theta \leftarrow \theta - \eta\, d_{Adam}$

η: learning rate · d_{Adam}: adaptive gradient direction



Actual training runs

Pale lines: training; dark lines: validation

Dotted lines: selected epochs

One action value: $(0.15 - 0.10)^2 = 0.0025$

The four experiments

Open the notebooks in order on the prepared course server.

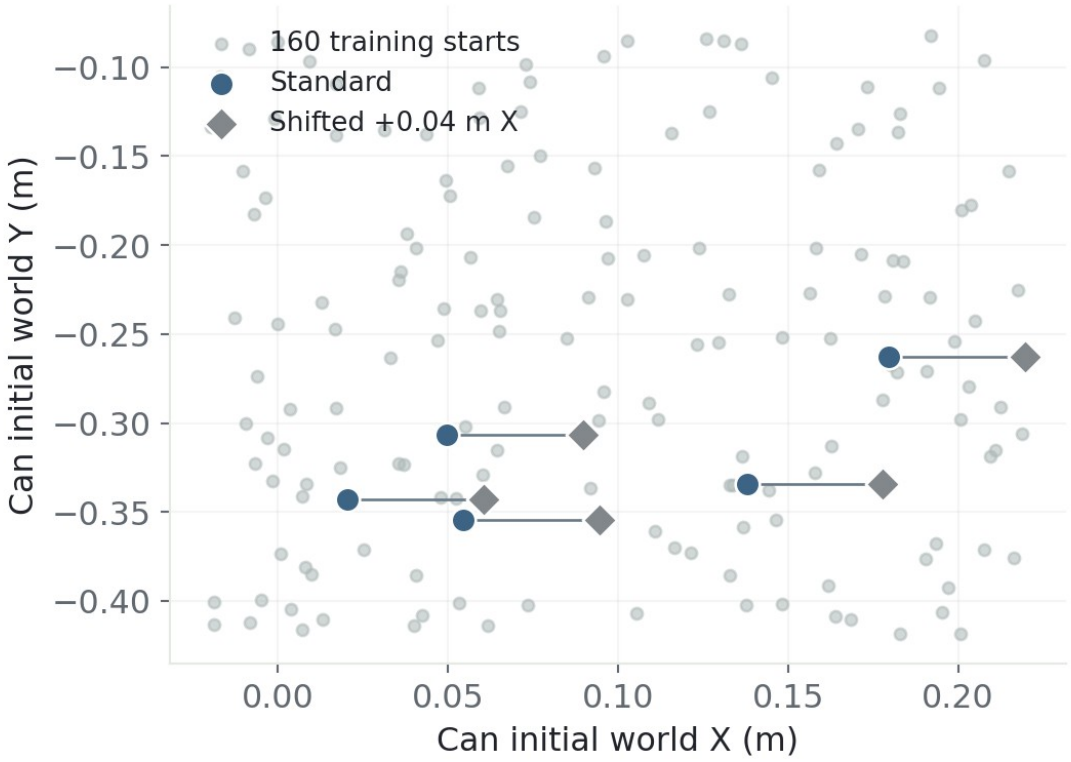
01	**Demonstrations** Play a demo; inspect observations and actions.	20 min	Q1
02	**Policy training** Train both rates; save models, curves and evaluations.	50–60 min	Q2–Q3
03	**Evaluation** Inspect the same saved cases and position shifts.	40–50 min	Q4
04	**Failure analysis** Select ≥3 steps; mark the first error; export evidence.	20–30 min	Q5

Compare $\eta = 0.001$ and $\eta = 0.0003$

Keep data, initial weights, architecture, seed 17, 50 epochs, batch size 256 and device fixed.

Evaluating a learned policy

Two models × (5 standard + 5 shifted starts) = 20 rollouts



Stable placements: one completed example

Learning rate	Standard	Shifted
0.001	2 / 5	3 / 5
0.0003	5 / 5	5 / 5

Paired source episode and seed

Can position shifted +0.04 m along world X

Five cases give a coarse estimate; a position shift is not a certified out-of-distribution test.

Analyzing a failure

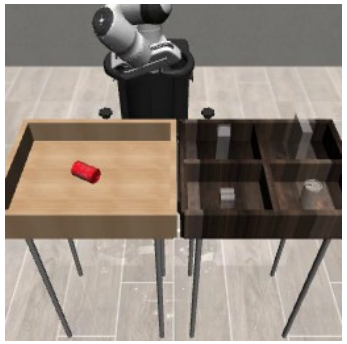
Locate the first visible error using frames and logged state.



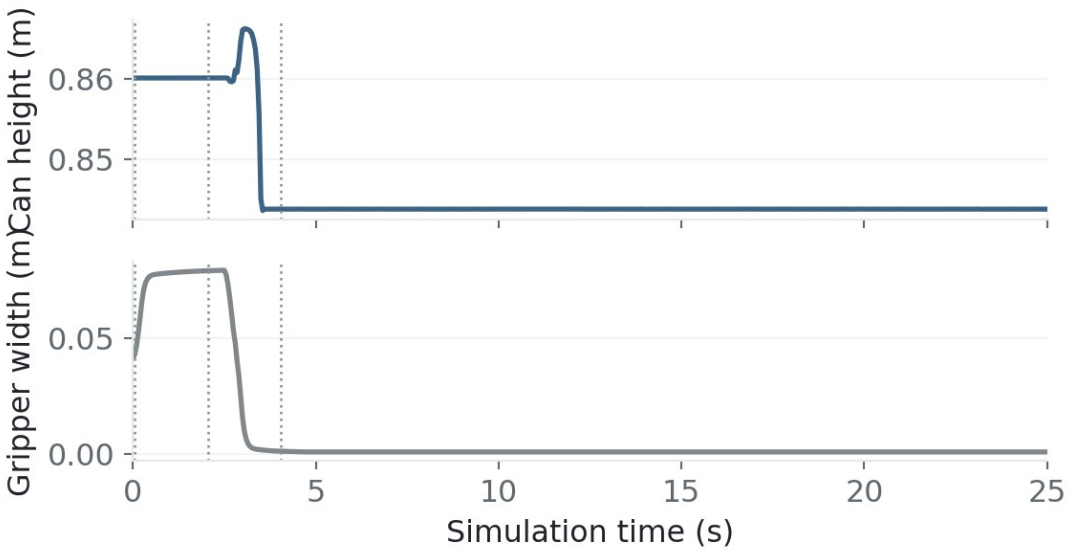
Step 1



Step 41



Step 81



Observed: the can remains in the source bin.

In Notebook 04, select ≥3 control steps and mark the first observable error.

Observation → hypothesis → proposed test

Check the offset between the requested state and the nearest recorded video frame.

Submission and assessment

Submit a 1–2 page PDF and the evidence ZIP

Answer Q1–Q5 in your own words.
Export the ZIP from Notebook 04.
Include both models and all outcomes.

The generated report is an evidence appendix.
Blank answers or no error marker remain DRAFT.

Assessment

Demonstration understanding	20
Training and experiment	25
Policy evaluation	30
Failure analysis and discussion	25

Success rate does not determine your grade.